

Total No. of Questions : 8]

SEAT No. :

PC2447

[Total No. of Pages : 2

[6354]-575

B.E. (Information Technology)

MOBILE COMPUTING

(2019 Pattern) (Elective-III) (Semester-VII) (414444 A)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Attempt Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data if necessary.*

Q1) a) Discuss the evolution of Wireless networks from 1 G to 3G , List out the advances in 2G, 2.5 G and 3 G Networks. **[10]**

b) Discuss in detail the advancements in 4G and 5G networks. **[7]**

OR

Q2) a) Discuss the evolutions of Mobile Communication Technologies in detail **[10]**

b) Outline the advantages of advancements in 4G networks over earlier generations. **[7]**

Q3) a) Illustrate the process of IP packet delivery with Encapsulation and tunnelling. **[10]**

b) Discuss the process of agent advertisement and discovery. **[7]**

OR

Q4) a) Illustrate with the diagram the process of data transfer to the mobile system using IPV4. **[10]**

b) Discuss in detail the problem of Triangular Routing and propose the solution for the same. **[7]**

P.T.O.

- Q5) a)** Explain TCP mechanisms with its operations, advantages and disadvantages. [10]
- b)** Compare I-TCP, Snooping TCP and M-TCP. [8]

OR

- Q6) a)** Discuss the problem of fast retransmit and timeout freezing in mobile transport layer and determine the proposed solution for the same. [10]
- b)** Illustrate how is WAP-based web access is different from Traditional Web access. [8]

- Q7) a)** Justify the constraints under which the operating system of a mobile device needs to operate. Conclude the special features that are required to be supported by a mobile OS, but are not present in traditional operating systems. [10]
- b)** Show with the diagram and explain Monolithic design versus microkernel design of an operating system. [8]

OR

- Q8) a)** Summarise the comparison of any three mobile operating systems that are widely used in mobile phones. [10]
- b)** Explain the architecture of mobile commerce framework and discuss advantages and disadvantages of mobile commerce. [8]

